BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: —

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. How many states of equilibrium:

(A) 1 (B) 2

(C) 3 (D) 4

Q2. On Celsius scale, 300 K = ?

(A) 27°C (B) 24°C

(C) 25°C (D) 26°C

Q3. One horse power is equal to:

(A) 746 W (B) 740 W

(C) 760 W (D) 700 W

Q4. Which one of the following unit is not a derived unit:

(A) Pascal (B) Kilogram

(C) Newton (D) Watt

Q5. The work done will be zero when the angle between the force and the distance is:

(A) 45° (B) 60°

(C) 90° (D) 180°

Q6. In solids heat is transferred by:

(A) Radiation (B) Conduction

(C) Convection (D) Absorption

Q7. A body has translatory motion if it moves along a:

(A) Straight line (B) Circle

(C) Line without rotation (D) Curved path

Q8. The number of forces that can be added by head to tail rule:

(A) 2 (B) 3

(C) 4 (D) Any number

Q9. The orbital speed of a low orbit satellite is:

(A) Zero (B) 8 ms^{-1}

(C) 800 ms^{-1} (D) 8000 ms^{-1}

Q10. The range of clinical thermometer is:

(A) 20°C – 42°C (B) 25°C – 42°C

(C) 30°C – 42°C (D) 35°C – 42°C

Q11. Which one of the following is the smallest quantity:

(A) 0.01 g (B) 2 mg

(C) 100 µg (D) 8000 ng

Q12. SI unit of pressure is Pascal, which is equal to:

(A) 10^4 Nm^{-2} (B) 1 Nm^{-2}

(C) 10^2 Nm^2 (D) 10^3 Nm^{-3}

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Differentiate between base quantities and derived quantities.

(ii) Define plasma physics.

(iii) Write down two rules to identify the significant figures.

(iv) Differentiate between vectors and scalars.

(v) Define position.

(vi) Write down mathematical form of 3rd equation of motion.

(vii) Write down two disadvantages of friction.

(viii) Differentiate between centripetal and centrifugal force.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) What is a solar cell? Write down its uses.

(ii) Why an energy saver lamp is better than an electric lamp?

(iii) Define power. Write down its SI unit.

(iv) Define force of gravitation.

(v) What is meant by artificial satellites? Write down their uses.

(vi) Why does the value of 'g' vary from place to place?

(vii) Write difference between like and unlike parallel forces.

(viii) Define centre of mass.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is meant by elastic limit?

(ii) Define density and write its unit.

(iii) Write four differences between solid and gas state of matter.

(iv) What is meant by internal energy of a body?

(v) Define specific heat capacity.

(vi) What is meant by transfer of heat? Write its modes.

(vii) Upon which factors radiation depend?

(viii) Why does land breeze blow in the night?

> Note: Attempt any TWO questions. (PART – II)

Q5.

(a) [1,3] Derive the second equation of motion with the help of speed-time graph.

(b) [5] How much centripetal force is needed to make a body of mass 0.5 kg to move in a circle of radius 50 cm with a speed 3 ms^{-1}?

Q6.

(a) [4] Explain any two states of equilibrium.

(b) [5] Calculate the value of 'g' at a height of 3600 km above the surface of the earth.

Q7.

(a) [4] Define and explain latent heat of fusion.

(b) A student presses her pain by her thumb with a force of 75 N. What would be the pressure under the thumb having contact area 1.5 cm^2?

GROUP – II

Objective Type

Time Allowed: 15 Min.	**Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Inertia depends upon:

(A) Force
(B) Net force
(C) Mass
(D) Velocity

Q2. The specific heat of glass is:

(A) 840 $Jkg^{-1} K^{-1}$
(B) 645 $Jkg^{-1} K^{-1}$
(C) 850 $Jkg^{-1} K^{-1}$
(D) 900 $Jkg^{-1} K^{-1}$

Q3. The value of 'g' on the surface of sun is:

(A) 8.87 ms^{-2}
(B) 25.94 ms^{-2}
(C) 274.2 ms^{-2}
(D) 9.8 ms^{-2}

Q4. Amount of a substance is measured in:

(A) Gram
(B) Kilogram
(C) Newton
(D) Mole

Q5. Rate of doing work is called:

(A) Energy
(B) Torque
(C) Power
(D) Momentum

Q6. Rate of flow of heat is:

(A) $Q \times t$
(B) $Q^2 \times t$
(C) Q^2/t
(D) Q/t

Q7. Which of the following is a vector quantity:

(A) Speed
(B) Distance
(C) Displacement
(D) Power

Q8. The unit of torque is:

(A) N
(B) Nm
(C) J
(D) Nm^2

Q9. Moon completes in one revolution around the earth in:

(A) 25.3 days
(B) 29.3 days
(C) 31.3 days
(D) 27.3 days

Q10. In gases heat is mainly transferred by:

(A) Molecular collision
(B) Conduction
(C) Convection
(D) Radiation

Q11. Least count of digital vernier callipers is:

(A) 0.01 mm
(B) 0.001 mm
(C) 0.1 mm
(D) 1 mm

Q12. In SI unit of stress is:

(A) Nm^{-1}
(B) Nm^{-2}
(C) Ns
(D) Nm

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) What are prefixes and give two examples.

(ii) Define least count of vernier callipers and write value of leat count of vernier callipers in centimetre.

(iii) Define plasma physics.

(iv) Differentiate between scalars and vectors quantities.

(v) What do you know about LIDAR gun?

(vi) Define uniform acceleration.

(vii) State Newton's Third Law of motion and give one example.

(viii) Write two methods of reducing friction.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Define stable equilibrium.

(ii) What is first condition for equilibrium?

(iii) Define clockwise and anti-clockwise moment.

(iv) Define geostationary orbit.

(v) What is meant by satellite and natural satellite?

(vi) How the value of 'g' varies with altitude?

(vii) Define biomass energy and geothermal energy.

(viii) What is meant by power and watt?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define density and also give its SI unit.

(ii) State Pascal's law.

(iii) What is meant by elastic unit?

(iv) Define volume thermal expansion.

(v) Convert 100°F into the temperature on Celsius scale.

(vi) Why conduction of heat does not take place in gases?

(vii) Why does sea breeze blow during the day?

(viii) Write down two uses of insulators.

> Note: Attempt any TWO questions. (PART – II)

Q5.

(a) [1,3] Draw speed time graph and derive second equation of motion $S = v_i t + 1/2\ at^2$.

(b) [5] A body has weight 20 N. How much force is required to move it vertically upward with an acceleration of $2ms^{-2}$?

Q6.

(a) [4] What is center of gravity? Explain how do you find the center of gravity of an irregular shaped thin lamina.

(b) [5] A block weighing 20 N is lifted 6 m vertically upward. Calculate the potential energy stored in it.

Q7.

(a) [4] Explain linear thermal expansion in solids.

(b) [5] The density of air is 1.3 kg^{-3}. Find the mass of air in a room measuring 8m × 5m × 4m.